RAJALAKSHMI ENGINEERING COLLEGE

Malathi M 251001142-2025-IT

M2

Started on	Friday, 31 July 2026, 3:30 PM
State	Finished
Completed on	Friday, 31 July 2026, 4:06 PM
Time taken	36 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
    int i=1;
    count++;
    int s=1;
    count++;
    while(s <= n)
    {
        count++;
        i++;
        count++;
         s += i;
         count++;
     }
     count++;
   printf("%d",count);
   return 0;

}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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M2

Started on	Friday, 31 July 2026, 3:52 PM
State	Finished
Completed on	Friday, 31 July 2026, 4:11 PM
Time taken	18 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
    if(n==1)
    {
      count++;
      //printf("*");
    }
    else
    {
        count++;
     for(int i=1; i<=n; i++)
     {
         count++;
       for(int j=1; j<=n; j++)
       {
           count++;
         // printf("*");
          count++;
         // printf("*");
         count++;
          break;

       }
       count++;
     }
     count++;
   }
   printf("%d",count);
   return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Malathi M 251001142-2025-IT

M2

Started on	Friday, 31 July 2026, 4:11 PM
State	Finished
Completed on	Friday, 31 July 2026, 4:37 PM
Time taken	25 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int num;
    scanf("%d",&num);
    int count=0;
    for (int i = 1; i <= num;++i)
    {
        count++;

        count++;//for entering into if condition
     if (num % i== 0)
        {
          count++;
         // printf("%d ", i);
        }

    }
    count++;

    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Malathi M 251001142-2025-IT

M2

Started on	Friday, 31 July 2026, 4:37 PM
State	Finished
Completed on	Friday, 31 July 2026, 7:12 PM
Time taken	2 hours 35 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c= 0;
    int count=0;
    count++;
    for(int i=n/2; i<n; i++){
      count++;
        for(int j=1; j<n; j = 2 * j){
          count++;
            for(int k=1; k<n; k = k * 2) {
               count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
 }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Malathi M 251001142-2025-IT

M2

Started on	Friday, 31 July 2026, 7:15 PM
State	Finished
Completed on	Friday, 31 July 2026, 7:32 PM
Time taken	17 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
     int rev = 0, remainder;
     int count=0;
    count++;
   count++;
   while (n != 0)
    {

        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n/= 10;
        count++;

    }
  count++;
  printf("%d",count);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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